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The efficacy and safety of dupilumab for the treatment of atopic dermatitis among Chinese patients in clinical practice: A single-center retrospective study

Ying Li[1,2] | Jiajing Lu[1,2] | Rongfen Chen[1,2] | Yu Wang[1,2] | Yangfeng Ding[1,2] | Shuang Xu[1,2] | Ying Zou[1,2] | Xuemei Yi[1,2] | Yuling Shi[1,2]

[1]Department of Dermatology, Shanghai Skin Disease Hospital, Tongji University School of Medicine, Shanghai, China

[2]Institute of Psoriasis, Tongji University School of Medicine, Shanghai, China

Correspondence
Yuling Shi and Xuemei Yi, Department of Dermatology, Shanghai Skin Disease Hospital, Tongji University School of Medicine, Shanghai, 200443, China.
Email: shiyuling1973@tongji.edu.cn and xuemei_yi@aliyun.com

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Abstract

Little real-work data regarding the efficacy and safety of dupilumab in the treatment of atopic dermatitis (AD) is available at present. To assess the efficacy and safety of dupilumab at 12 weeks in the treatment of AD in clinical routine clinical practice. A retrospective, single-centre study of adult patients with moderate to severe AD treated with dupilumab for 12 weeks in China. In total, 60 patients (48 male, 12 female; mean age: 53.2 ± 15.6) were enrolled in this retrospective study. These patients exhibited a mean AD disease course of 10.6 years (6.0), 30% exhibited a family history of allergies, and 31 (51.7%) had one or more allergic comorbidities. Following dupilumab treatment for 12 weeks, 83.3% and 42% of patients had achieved EASI-50 and EASI-75, respectively. Overall, adverse events (AEs) were reported by 15% of patients, with the most common being conjunctivitis, injection site reactions, and herpes simplex virus infections. Laboratory testing after 12 weeks revealed pronounced decreases in both circulating eosinophil counts (from 0.6 (0.1–2.8) to 0.3 (0.1–9.7) 10^9/L) and total IgE concentrations (from 327 (2.46–2500) to 230 (47.6–2200) U/ml) in these patients. These real-world data reaffirm the safety and efficacy of dupilumab as a treatment for moderate-to-severe AD among Chinese patients in clinical practice.

KEYWORDS
atopic dermatitis, dupilumab, efficacy, real word, safety

1 | INTRODUCTION

Atopic dermatitis (AD) is a chronic inflammatory disease of the skin that affects an estimated 15%–30% of children and 2%–10% of adults, resulting in potentially severe pruritus.[1] The pathogenic etiology of AD is complex and multifactorial, with interleukin (IL)-4 and IL-13 playing important proinflammatory roles, resulting in the impaired expression of epidermal structural proteins and the consequent development of inflamed pruritic lesions.[2] The effective treatment of moderate-to-severe AD remains challenging, with several agents such as JAK inhibitors and biologics targeting IL-4 and IL-13 having emerged as promising therapeutic modalities for affected patients. Duplizumab has been approved for the treatment of moderate-to-severe AD, with the humanized monoclonal anti-IL-4 receptor α antibody dupilumab having exhibited good efficacy and reasonable safety in clinical trials attributable to its ability to disrupt IL-4 and IL-13 signaling.[3]

However, clinical trials are subject to several limitations attributable to their strict exclusion and inclusion criteria such that real-world

Ying Li, Jiajing Lu, Rongfen Chen, and Yu Wang contributed equally to this work.

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clinical data often differ substantially from clinical trial results. Dupilumab also showed good efficacy, safety, and tolerability on some special AD patients such as HIV-positive patients and elderly patients,[4,5] which were excluded from clinical trials. As dupilumab was approved for use in China last year, little real-world clinical data pertaining to its safety and efficacy in a Chinese patient population is available at present, and the publication of such data is important given that it has the potential to better inform treatment efforts in clinical practice.

2 | MATERIAL AND METHODS

2.1 | Study design

This was a retrospective, single-center, comparative study of all patients who had been hospitalized for treatment with dupilumab at Shanghai Skin Disease Hospital between September 2020 and October 2021. All patients with moderate-to-severe AD that were 18 years of age or older were treated subcutaneously with dupilumab at the standard recommended dose (600 mg at Week 0, followed by 300 mg every 2 weeks). Patients were permitted to use moisturizers, topical corticosteroids, and topical calcineurin inhibitors as indicated. Patient baseline data pertaining to age, gender, comorbidities, and prior AD treatments were recorded. Patient responses to treatment were assessed by comparing Eczema Area and Severity Index (EASI, 0–72), Dermatology Life Quality Index (DLQI, 0–30), and Numeric Rating Scale (NRS)-itch (0–10) scores at baseline to those after 12 weeks of treatment. All adverse events (AEs) were also recorded for analysis. All patients provided informed consent to participate the study. The ethics committees of Shanghai Skin Disease Hospital approved the present study.

2.2 | Statistical analysis

Normally distributed data are given as mean ± standard error, while skewed data are given as medians and ranges. Data were compared via Student's *t*-test or Mann–Whitney *U* tests using SPSS 26.0 (IBM SPSS Statistics Base), with $p < 0.05$ as the significance threshold.

3 | RESULTS

3.1 | Baseline characteristics

Patient baseline demographic and clinical characteristics are summarized in Table 1. In total, 60 patients (48 male and 12 female) were included in this study, with a mean age of 53.2 ± 15.6. These patients exhibited a mean AD disease course of 10.6 ± 6.0 years, 30% exhibited a family history of allergies, and 31 (51.7%) had one or more allergic comorbidities. The most commonly reported allergic diseases in this patient population included allergic rhinitis (43.3%), allergic asthma (16.7%), and food allergies (10%), while the most common comorbidities included hypertension and type 2 diabetes mellitus, with a subset of patients exhibiting thyroiditis or epilepsy. All patients had undergone prior topical and systemic treatments for AD, with 36.7% having received compound glycyrrhizin, 16.7% having been treated with systemic glucocorticoids, and 15% having undergone phototherapy. Moreover, 25% of these patients had received other therapies, including traditional Chinese medicine (13.3%), tripterygium glycosides (10%), methotrexate (3%), and baricitinib (2%).

3.2 | Dupilumab efficacy, safety and clinical findings

Following dupilumab treatment for 12 weeks, 83.3% and 42% of patients had achieved 50% and 75% improvements in EASI scores, respectively (Figure 1, Table 2). At baseline, the mean EASI score was 29.5 ± 4.3, while the mean baseline DLQI score was 12.7 ± 3.2, and the mean baseline NRS-itch score was 7.5 ± 1.2 in this patient cohort,

TABLE 1 Baseline demographics and clinical characteristics of dupilumab for AD patients in the study ($N = 60$)

Characteristic	Mean (SD) or *N* (%)
Age, year	53.2 (15.6)
Sex, female	12 (20)
BMI (kg/m^2)	23.5 (2.6)
Disease duration, year	10.6 (6.0)
Family history of allergy	18 (30)
Allergic comorbidities	
Allergy history	42 (51.7)
Allergic rhinitis	26 (43.3)
Asthma	6 (16.7)
Food allergy	4 (10)
Comorbidities	
Hypertension	19 (31.7)
T2DM	6 (10)
Thyroiditis	2 (3)
Epilepsy	1 (2)
Chronic hepatitis B infection	1 (2)
Previous treatment	
Topical therapies	60 (100)
Antihistamines	60 (100)
Compound glycyrrhizin	22 (36.7)
Systemic corticosteroids	10 (16.7)
Phototherapy	9 (15)
Others	15 (25)
Traditional Chinese medicine	8 (13.3)
Tripterygium glycosides	3 (10)
Methotrexate	2 (3)
Baritinib	1 (2)

Abbreviations: AD, atopic dermatitis; BMI, body mass index; T2DM, type 2 diabetes.

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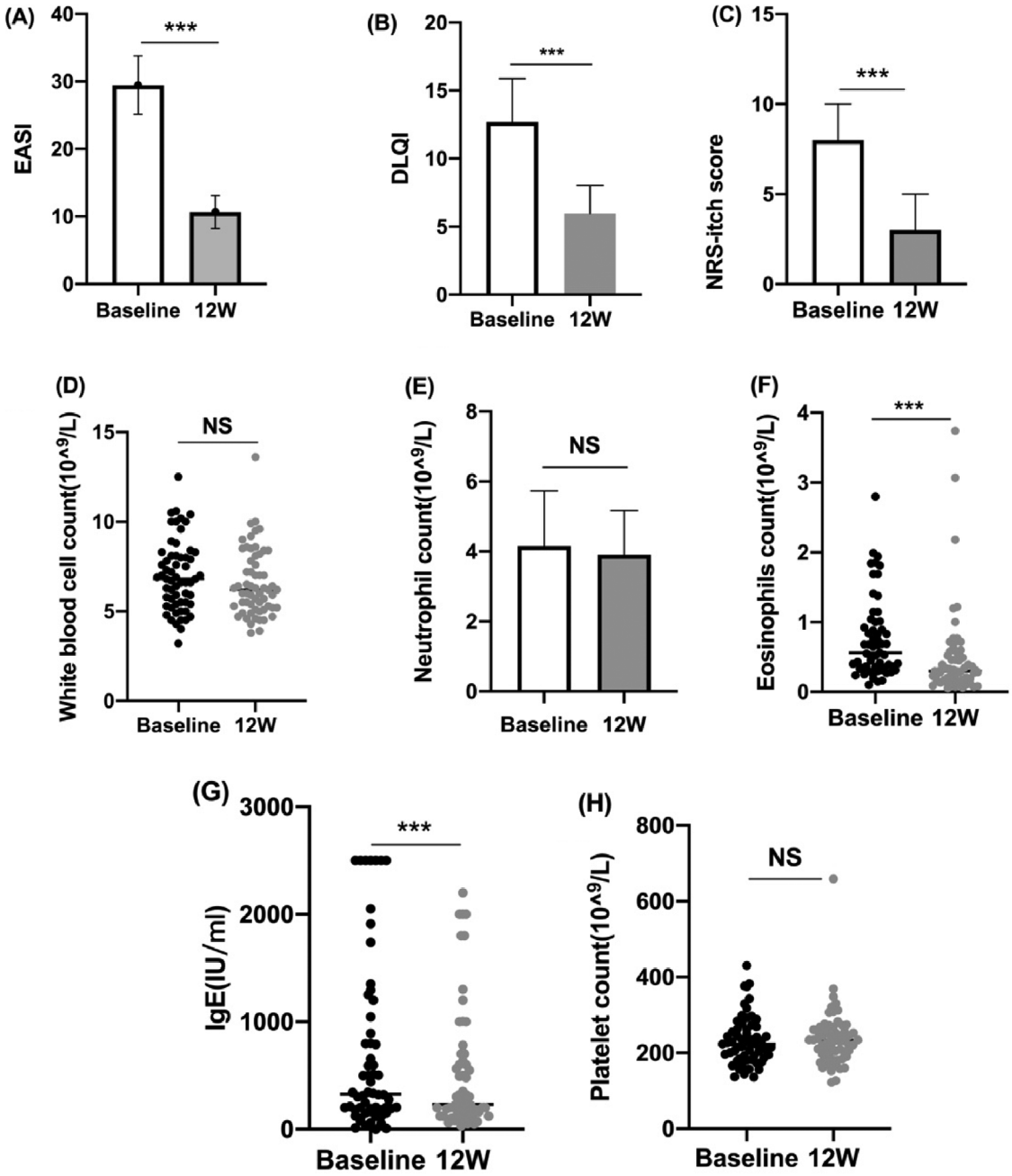

FIGURE 1 Patient characteristics and laboratory results of dupilumab for AD patients at baseline and follow-up after 12 week. AD, atopic dermatitis; DLQI, Dermatology Life Quality Index; EASI, Eczema Area and Severity Index; IgE, immunoglobulin E; NRS-itch score, Numeric Rating Scale (NRS)-itch score; ***$p < 0.001$ compared with the respective value at baseline; NS, there was no significance compared with the respective value at baseline

with these scores declining rapidly following the administration of dupilumab to 10.7 ± 2.4, 6.0 ± 2.1, and 3.3 ± 1.1, respectively. Overall, AEs were reported by 15% of patients, with the most common being conjunctivitis, injection site reactions, and herpes simplex virus infections. Other rare adverse reactions included fever, aggravated itching, increased rash severity, headache, and arthralgia (Table 3).

Laboratory testing after 12 weeks revealed pronounced decreases in both circulating eosinophil counts (from 0.6 (0.1–2.8) to 0.3 (0.1–9.7) 10^9/L) and total IgE concentrations (from 327 (2.46–2500) to 230 (47.6–2200) U/ml) in these patients, while no significant changes in white blood cell, neutrophil and platelet counts were observed (Figure 1, Table 1).

4 | DISCUSSION

In summary, this retrospective observational study indicated that dupilumab is a safe and effective treatment for moderate-to-severe AD in an adult Chinese patient population. We found that 86.7% and 42% of patients achieved EASI-50 and EASI-75 at 12 weeks of

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TABLE 2 Patient characteristics and laboratory results of dupilumab for AD patients at baseline and follow-up after 12 week

	Baseline	12 weeks
EASI score	29.5 ± 4.3	10.7 ± 2.4
DLQI score	12.7 ± 3.2	6.0 ± 2.1
NRS-itch score	7.5 ± 1.2	3.3 ± 1.1
White blood cell count (10^9/L)	6.8 (3.2–12.5)	6.2 (3.8–13.6)
Neutrophil count (10^9/L)	4.2 ± 1.6	3.9 ± 1.3
Eosinophils count (10^9/L)	0.6 (0.1–2.8)	0.3 (0.1–9.7)
Platelet count (10^9/L)	223 (136–430)	233 (122–659)
IgE (IU/ml)	327 (2.46–2500)	230 (47.6–2200)

Note: Normally distributed data are shown as mean ± standard error, skewed data are shown as median (min–max).
Abbreviations: AD, atopic dermatitis; DLQI, Dermatology Life Quality Index; EASI, Eczema Area and Severity Index; NRS-itch score, Numeric Rating Scale (NRS)-itch score.

TABLE 3 Adverse events reported of dupilumab for AD patients

	N (%)
Adverse events	9 (15)
Conjunctivitis	6 (10)
Injection site reaction	3 (5)
Herpes simplex virus	3 (5)
Fever	2 (3)
Aggravated itching	2 (3)
Increased rash	2 (3)
Headache	1 (1)
Arthralgia	1 (1)

Abbreviation: AD, atopic dermatitis.

treatment, respectively, aligning well with results from prior clinical trials (EASI-50: 65%–85% and EASI-75: 44%–69%).[6–8] Our data are also consistent with other real-world studies, with Olesen et al.[9] having found that 63.3% (19/30) of their patients achieved EASI-75 following a 3-month treatment course. Similarly, Faiz et al.[10] conducted a real-world multi-center retrospective analysis demonstrating that 48.8% (40/82) of patients with recorded EASI scores achieved EASI-75 or greater within 3 months.

The safety profile observed for dupilumab in this study was consistent with data from previous clinical trials and real-world studies.[11,12] Conjunctivitis was the most commonly reported AE, with reported incidence rates in clinical trials ranging from 5% to 28%[6–8] and rates of 6% to 62% having been reported in real-life studies.[13–15] At present, how dupilumab induces conjunctivitis is poorly understood. Potential mechanisms involve *Demodex* colonization, which can result in increases in eosinophil counts and IL-17-mediated inflammation.[16] Increased levels of eosinophilic factors such as eotaxin-1 and eotaxin-2, which are implicated in eosinophil recruitment, have also been observed in mucus and tear samples from atopic keratoconjunctivitis patients.[17] We also observed significant reductions in serum IgE levels in our patients, consistent with prior reports,[9,10,18] likely as dupilumab can disrupt IL-4 and IL-13 signaling, which drives IgE production.[19] In contrast to prior studies, however, we found that eosinophil counts were decreased in our patients following a 12-week treatment course, whereas others have reported no changes in eosinophil counts after treatment for 16 weeks.[6–8,15] Some real-world studies have even reported that the proportion of patients treated with dupilumab exhibiting eosinophilia within 6 months (57%) or 16 weeks (43%) were significantly increased relative to baseline levels (33% and 31%, respectively).[10,20]

5 | CONCLUSION

Our results suggest that dupilumab exhibits a good short-term efficacy and safety profile in a real-world clinical population of Chinese adults with moderate-to-severe AD. However, future studies will be necessary to more fully establish the clinical responses of a larger population of patients to long-term dupilumab.

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CONFLICT OF INTEREST

The authors declare no potential conflict of interest.

AUTHOR CONTRIBUTIONS

All authors contributed to the design and implementation of the research. Ying Li contributed to the writing. Jiajing Lu contributed to the analysis of the results of the manuscript. Rongfen Chen, Wang Yu and Shuang Xu contributed to collect data. Yang feng Ding, Ying Zou, Xuemei Yi and Yuling Shi contributed to the supervision of the final manuscript.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ORCID

Ying Li https://orcid.org/0000-0002-4041-2646
Jiajing Lu https://orcid.org/0000-0002-0827-0596
Yuling Shi https://orcid.org/0000-0002-1273-7881

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